Muhammad Waqar

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LinkedIn | Portfolio

Summary

AI engineer with 7+ years of building machine learning and generative AI systems that run in production. At Softoo I lead the agentic AI and RAG work, including an audit platform that automates SAMA, ISO and BCM compliance reviews. Most of my recent work is multi-agent workflows in LangGraph, RAG, text-to-SQL, and running open-source LLMs on-premise alongside Azure OpenAI for clients who can't send their data to the cloud. Before Softoo I led ML teams working on computer vision and NLP, and I have a paper published at IEEE ICRAI.

Experience

Softoo May 2025 – Present

AI Lead Engineer (Generative AI, RAG Systems, AI Agents) *Islamabad, Pakistan*

- Lead the technical side of our agentic AI and RAG products for enterprise automation, from architecture through to production releases.
- Designed multi-agent workflows in LangGraph and LangChain where agents plan multi-step tasks and call tools, instead of following hard-coded rule pipelines.
- Built RAG pipelines (vector database + semantic search) that answer from the client's own knowledge base and show the source behind each answer.
- Run SQLCoder, Llama and Code Llama on-premise for clients whose data can't leave their network. This also brought inference costs down compared to paying for hosted APIs.
- Led the build of our AI audit platform for BCM, SAMA and ISO (more under Projects): risk and control generation, control testing and the final audit report.
- Mentor the junior engineers and set up how the team builds and governs agents in production.

Forbmax (Pvt) Ltd May 2023 – May 2025

AI Team Lead (Computer Vision, NLP, Data Science) *Islamabad, Pakistan*

- Led a team of ML engineers working across computer vision, NLP and data science. Set the technical direction and did most of the core architecture myself.
- Rebuilt our model pipelines end to end (preprocessing, training, deployment) with automated workflows, which made the team about 40% more efficient.
- Fine-tuned transformers, YOLO and vision transformers on client data with transfer learning, since off-the-shelf models rarely reached the accuracy clients needed.
- Built an active-learning annotation tool that cut manual labelling by 70% without hurting label quality.
- Pushed the team to ship models as APIs and microservices instead of one-off scripts, which made them much easier to plug into other products.

Horizon Tech Services Jun 2020 – Apr 2023

Senior AI Engineer *Islamabad, Pakistan*

- Ran ML projects from data collection through to deployment and mentored the engineers on the team.
- Built preprocessing pipelines that automated feature extraction and data augmentation, which improved model results and saved a lot of manual data work.
- Spent much of my time fine-tuning object detection and tracking models for production use, improving recognition and tracking accuracy.

Lampro Mellon IT Solutions (Pvt) Ltd Aug 2019 – May 2020

Python Developer *Lahore, Pakistan*

- Wrote Python software components, mainly around verification and high-level design.
- Implemented algorithms and data-processing modules, and wrote scripts to automate verification that was previously done by hand.

Projects

Enterprise Compliance AI Agent Platform (Softoo) | *Agentic AI, RAG, LangGraph* Sep 2026

- Led the design of an agentic audit platform for BCM, SAMA and ISO compliance, including maturity assessments of policies, contracts and internal documents.
- The LangGraph/LangChain workflow covers the whole audit: planning the engagement, building the risk and control matrix, running control tests and writing the report.
- Built the RAG layer (vector store + semantic search) so findings are based on the client's actual documents rather than the model's general knowledge.
- Sensitive data is handled by self-hosted Llama and DeepSeek models; Azure OpenAI/OpenAI is only used where the client's policy allows it.
- Added guardrails, audit logs and policy checks. Auditors can follow any finding back to the document it came from. Live at yanipro.ai.

ActPilot – AI Browser Agent | *Chrome Extension, Azure OpenAI, Groq* Aug 2026

- Chrome/Edge extension (Manifest V3) that reads the page text plus a screenshot, then actually does things on the page, like filling fields, clicking buttons and opening links, based on what the user types.
- The backend returns one JSON response per request, and the model can only choose from a fixed set of actions on elements it was shown. This stops it from inventing buttons that don't exist.
- Hard safety rule: it will fill in a form but won't submit, pay, delete or send anything unless the user explicitly asks.
- Uses Azure OpenAI's vision model and switches to Groq automatically when Azure hits rate limits.
- Also draws charts and exports chats as branded PDFs, with rate limiting built in. Published on the Microsoft Edge Add-ons store.

Private Knowledge Assistant | *Agentic RAG, Text-to-SQL, Web RAG, Azure OpenAI* Jan 2026 – Oct 2026

- A private assistant that answers from your own documents, web pages and live databases in the same chat. A tool-calling agent decides for each question whether to search documents, query a database, do both, or just reply.
- Document and web RAG over PDF, DOCX, TXT, images (OCR for scanned pages) and URLs. Every user's and every chat's data is kept separate at the vector-store level.
- Text-to-SQL/NoSQL for PostgreSQL, MySQL, SQL Server, SQLite and MongoDB, using schema-aware RAG with few-shot examples taken from foreign keys and from queries that worked before.
- Databases are strictly read-only: five layers of checks per engine (blocked keywords, forced rollback, statement timeouts, row limits), and credentials are encrypted with Fernet.
- Answers cite the exact source (page, URL or the SQL that ran), stream over SSE and can be shown as charts. JWT login. Live at knowledgeassistant-ai.vercel.app.

Private Voice Assistant | *Twilio, Whisper STT, Orpheus TTS* Aug 2025

- AI phone helpline that any business can configure at runtime. Works on real calls through Twilio and in a browser hold-to-talk demo, both on the same Groq pipeline.
- Got a full turn (speech-to-text, LLM, text-to-speech) down to 2–2.7 seconds by warming up connections and caching greetings.
- Detects the caller's language from their speech and answers in that language.
- Used HTTP polling with Twilio rather than WebSocket media streams so the whole thing could run serverless. Live at voiceassistant-ai.vercel.app.

WellB – AI-Powered Hotel Management System | *RAG, Llama, DeepSeek, OpenAI* Feb 2025

- Hotel assistant built on RAG, using local Llama and DeepSeek models alongside OpenAI GPT.
- Users can book or change bookings in plain language; it checks for conflicts and reschedules automatically, with DeepSeek handling the multi-step reasoning.
- Emergency alerts with real-time notifications, prioritised by the model.

- Contact details are extracted and saved by a locally hosted Llama model, so guest data stays on our own servers.

IntelliReport – AI-Powered Report Generation | *Python, NLP, Transformers, FastAPI* Sep 2024

- Report generator that cleans up text, pulls out the main topics and summarises each one, using BERT-based models.
- Mixes extractive and abstractive summarisation so summaries stay short without dropping key points.
- Outputs Markdown/HTML for the frontend and a downloadable PDF.
- Runs as a FastAPI service with batch jobs and live progress updates for long documents.

Smart Media Monitoring Application | *Python, PyTorch* Aug 2023

- Led a media monitoring system that brings speech-to-text, sentiment analysis, summaries, topic modelling, face recognition and OCR together in one pipeline.
- Trained and tested on both Urdu and English data.
- OCR pipeline for PDFs, images and tables in Urdu, English and Arabic.
- Face recognition to identify and follow people across media content.

Smart Surveillance System | *Python, PyTorch, TensorFlow* Dec 2022

- Led the AI side of a security monitoring system running on live and IP camera streams.
- Real-time object detection and tracking with low latency.
- Number plate recognition (ANPR) for identifying vehicles, with support for plates in more than one language.
- Geofencing: virtual boundaries drawn inside camera views for automatic area monitoring.

AAQAB – Satellite Imagery Intelligence | *Python, YOLO, PyTorch* Feb 2022

- Led object detection on high-resolution satellite images using YOLO and ResNet.
- The images were too large to process whole, so we tiled them by latitude and longitude and spread the work across GPUs.
- Used SAHI sliding-window inference, which clearly improved detection and counting of small objects.

Technical Skills

Generative AI & Agents: LLM Orchestration, Agentic Tool-Calling, RAG Pipelines, LangChain, LangGraph, Prompt Engineering, OpenAI SDK, Azure OpenAI, GCP/Vertex AI, AWS Bedrock, Groq, Open-Source LLMs (Llama, Mistral, DeepSeek), On-Prem LLM Hosting

ML & Computer Vision: PyTorch, TensorFlow, Transformers, YOLO, scikit-learn, Keras, OpenCV, AWS SageMaker

Voice & Speech AI: Groq Whisper STT, Azure Speech, Text-to-Speech, SeamlessM4T (Meta), Twilio Voice, Real-time Audio Pipelines

Data & Retrieval: PostgreSQL, MySQL, SQL Server, MongoDB, Text-to-SQL, SQLAlchemy, Alembic Migrations

Full-Stack & DevOps: FastAPI, React, TypeScript, Docker, AWS Lambda, JWT Auth, CI/CD (pytest/vitest), Vercel, Git

Leadership & Practice: Team Leadership, System Architecture, Security Engineering, Mentorship, Test-Driven Development

Education

University of Engineering and Technology (UET) Oct 2015 – Aug 2019
Bachelor of Science in Computer Engineering *Taxila, Pakistan*

Publications

Meter Digit Recognition Via Faster R-CNN – Muhammad Waqar, M. Waris, et al., M. Yousaf. *IEEE International Conference on Robotics and Automation in Industry (ICRAI)*, Oct 2019. DOI: 10.1109/ICRAI47710.2019.8967357.